

DDoS Incident Report

Report Period	Sep 26, 2026 20:44 - Sep 26, 2026 20:50
Generated At	2026-09-26 20:50:36
Classification	ML: Isolation Forest + Random Forest

1. Executive Summary

Metric	Value	Traffic Summary	Count
Total Threats Mitigated	26	Total Flows Observed	16421
High Priority	2	Total Packets Analyzed	6498524636
Low Priority	24	Malicious Packets	6483239156
Manual Releases	0	True Negatives Passed	10811
Manual Blocks	0		
ML Metrics	Value	Attack Vector	Count
False Positives	45	ICMP Flood	0
FP Rate	0.33%	SYN Flood	20
IF Precision	99.37%	UDP Flood	1
IF Recall	95.42%	Uncertain	5
IF F1-Score	97.36%		
IF Accuracy	98.16%	Action	Count
		Quarantined	4
		Time Ban	4
		Blackhole	0
		Released	3

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.37%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	95.42%	Share of actual attacks successfully flagged
F1-Score	97.36%	Balanced measure combining Precision and Recall
Accuracy	98.16%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.33%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	4.58%	Actual attacks that went undetected
True Positive Rate (TPR)	95.42%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.67%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 5559	FN 267
Actual: Normal	FP 35	TN 10544

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	99.95%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	99.97%	Balanced measure combining Precision and Recall
Accuracy	99.95%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	8.57%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	91.43%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	5559	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	0

2c. Hybrid Cascaded

Metric	Value	Description
Precision	99.37%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	95.42%	Share of actual attacks successfully flagged
F1-Score	97.36%	Balanced measure combining Precision and Recall
Accuracy	98.16%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.33%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	4.58%	Actual attacks that went undetected
True Positive Rate (TPR)	95.42%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.67%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 5559	FN 267
Actual: Normal	FP 35	TN 10544

2d. Response Latency

Metric	Value	Description
Detection Time	300.9 ms	Average time taken to detect an attack
Mitigation Response Time	3.6 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	0.87%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	22.21%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	19.93%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.16	5	4	2026-09-26 20:47:31	2026-09-26 20:50:11
10.0.0.19	5	4	2026-09-26 20:47:30	2026-09-26 20:50:12
10.0.0.17	4	3	2026-09-26 20:47:38	2026-09-26 20:50:20
10.0.0.18	4	3	2026-09-26 20:47:29	2026-09-26 20:50:10
10.0.0.3	1	1	2026-09-26 20:46:46	2026-09-26 20:46:56

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-26 20:46:56	10.0.0.3	DDoS	Uncertain	58.9%	Low	Released
2026-09-26 20:47:29	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 20:47:30	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 20:47:31	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 20:47:38	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 20:47:40	10.0.0.18	DDoS	SYN Flood	92.5%	Critical	Time Ban (30s)
2026-09-26 20:47:40	10.0.0.18	DDoS	SYN Flood	92.5%	Critical	Time Ban
2026-09-26 20:47:41	10.0.0.19	DDoS	SYN Flood	93.4%	Critical	Time Ban (30s)
2026-09-26 20:47:41	10.0.0.16	DDoS	SYN Flood	93.1%	Critical	Time Ban (30s)
2026-09-26 20:47:42	10.0.0.19	DDoS	SYN Flood	93.4%	Critical	Time Ban
2026-09-26 20:47:42	10.0.0.16	DDoS	SYN Flood	93.1%	Critical	Time Ban
2026-09-26 20:47:49	10.0.0.17	DDoS	SYN Flood	90.9%	Critical	Time Ban (30s)
2026-09-26 20:47:49	10.0.0.17	DDoS	SYN Flood	90.9%	Critical	Time Ban
2026-09-26 20:48:11	10.0.0.19	DDoS	SYN Flood	93.4%	Critical	Released
2026-09-26 20:48:11	10.0.0.16	DDoS	SYN Flood	93.3%	High	Released
2026-09-26 20:48:22	10.0.0.16	DDoS	SYN Flood	93.3%	Critical	Time Ban (1m)
2026-09-26 20:48:22	10.0.0.19	DDoS	SYN Flood	93.4%	Critical	Time Ban (1m)
2026-09-26 20:48:40	10.0.0.18	DDoS	SYN Flood	93.2%	Critical	Time Ban (1m)
2026-09-26 20:48:49	10.0.0.17	DDoS	SYN Flood	90.9%	Critical	Time Ban (1m)
2026-09-26 20:49:22	10.0.0.16	DDoS	SYN Flood	94.0%	Critical	Time Ban (2m)
2026-09-26 20:49:22	10.0.0.19	DDoS	SYN Flood	95.9%	Critical	Time Ban (2m)
2026-09-26 20:50:01	10.0.0.9	DDoS	UDP Flood	92.4%	High	Time Ban (30s)
2026-09-26 20:50:10	10.0.0.18	DDoS	SYN Flood	93.4%	Critical	Time Ban (2m)
2026-09-26 20:50:11	10.0.0.16	DDoS	SYN Flood	90.9%	Critical	Time Ban (5m)
2026-09-26 20:50:12	10.0.0.19	DDoS	SYN Flood	90.9%	Critical	Time Ban (5m)
2026-09-26 20:50:20	10.0.0.17	DDoS	SYN Flood	93.1%	Critical	Time Ban (2m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.17	SYN Flood	0.6266	93.1%	Critical	Phase 2	1	10s	2026-09-26 20:50:20	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6521	90.9%	Critical	Phase 2	1	11s	2026-09-26 20:50:12	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6369	90.9%	Critical	Phase 2	1	11s	2026-09-26 20:50:11	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.18	SYN Flood	0.6462	93.4%	Critical	Phase 2	1	11s	2026-09-26 20:50:10	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6383	95.9%	Critical	Phase 2	4	1m 11s	2026-09-26 20:49:22	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6132	94.0%	Critical	Phase 2	4	1m 11s	2026-09-26 20:49:22	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6194	90.9%	Critical	Phase 2	3	1m 11s	2026-09-26 20:48:49	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6476	93.2%	Critical	Phase 2	3	1m 11s	2026-09-26 20:48:40	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6451	93.4%	Critical	Phase 2	3	11s	2026-09-26 20:48:22	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6299	93.3%	Critical	Phase 2	3	11s	2026-09-26 20:48:22	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6246	90.9%	Critical	Phase 2	2	41s	2026-09-26 20:48:19	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6022	93.3%	High	Phase 2	1	40s	2026-09-26 20:48:11	Ban Expired
10.0.0.19	SYN Flood	0.6173	93.4%	Critical	Phase 2	1	41s	2026-09-26 20:48:11	Ban Expired
10.0.0.18	SYN Flood	0.6431	93.2%	Critical	Phase 2	2	41s	2026-09-26 20:48:10	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6264	90.9%	Critical	Phase 2	1	11s	2026-09-26 20:47:49	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6321	93.1%	Critical	Phase 2	1	10s	2026-09-26 20:47:41	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6499	93.4%	Critical	Phase 2	1	11s	2026-09-26 20:47:41	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6462	92.5%	Critical	Phase 2	1	11s	2026-09-26 20:47:40	Attack persisted - escalating to time ban
10.0.0.3	Uncertain	0.6085	58.9%	Low	Phase 1	1	10s	2026-09-26 20:46:56	Attack Stopped

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE